

Online Appendix

Before the Labor Market or Inside It?

Family Background, Earnings Dynamics, and the Return to College

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October 2026

This appendix has three parts. Part I reports supplementary results in the order of the paper: the data, the earnings process, the attenuation schedule, the premium, the expansion design, and the decomposition. Part II states and proves the formal results the paper uses. Part III is a guide to the replication files: which script produces each table and figure, how to run the pipeline, and how to obtain the data. Every table in Part I is written by a script from the estimation output, and the repository carries the checks that compare every number printed in [the paper](#) and in this appendix with that output.

Table A1: Where to find the material behind each section of the paper.

Section of the paper	Supplementary results	Formal results
2. Data	Section A	
3. The earnings process and family background	Section B	Part II, minimum-distance estimator
4. What the process implies for measurement	Section C	Part II, attenuation schedule
5. The level premium	Section D	
6. The college channel: the 1999 expansion	Section E	
7. Decomposing the premium	Section F	Part II, content of the direct term

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Part I. Supplementary results

A Data and sample

A.1 Income variables by wave

The survey’s measure of personal labor income changed definition in 2014 (Section 2.1 of the paper). Table A2 lists the variable used in each wave. The earnings outcomes of the paper use the waves in which the variable covers jobs only, 2014 to 2022. For the two estimates that use a seven-wave panel, a measure that covers jobs only is reconstructed for 2010 and 2012 from the components that the questionnaire records. For 2010 these are the monthly wage of the main job (annualized), the annual bonus, self-employment income, and income from secondary jobs; for 2012, the annual after-tax income of each job, the net income of each business, income from part-time work, and income from work for other agricultural households. Among all respondents aged 22 to 55 with positive income, the variance of log income under the survey’s own variable of Table A2 is 2.26 in the 2010 wave and 1.37 in the 2012 wave, against 0.87, 1.04, 0.85, 0.93, and 0.98 in the five waves from 2014 to 2022 ([unified/code/u17_income_variance_by_wave.py](#)).

Table A2: The CFPS income variable used in each wave.

Wave	Variable	Concept
2010	<code>income</code>	jobs, transfers and asset income together
2012	<code>income_adj</code>	jobs, transfers and asset income together
2014, 2016, 2018	<code>income</code>	jobs only
2020, 2022	<code>emp_income</code>	jobs only

A.2 Sample screens by family background

Table A3 follows children with a valid retrospective report of parental occupation through the screens of Section 2.2 of the paper. The screens pass advantaged and less advantaged children at nearly the same rate, so selection into the sample is unlikely to account for the difference between the two groups. Before any earnings screen, positive labor income appears in 55.6 percent of the person-waves of advantaged and 49.7 percent of the person-waves of less advantaged children at ages 22 to 55.

Table A3: Sample screens by family background.

Screen	Children		Percent of first row	
	Advantaged	Less advantaged	Advantaged	Less advantaged
Retrospective report, family income of the parental household, observed at ages 22–55 in a 2014–2022 wave	1,661	9,013	100.0	100.0
Trimming and at least two waves of positive labor income at ages 22–55 (the sample of the paper)	755	3,781	45.5	42.0
Of these, at least three waves (Sections 3 and 4 of the paper)	420	2,103	25.3	23.3
Of the sample, at least two waves at ages 30 and over (level of earnings, Sections 5 to 7 of the paper)	453	2,321	27.3	25.8

Notes: Advantaged children have at least one parent in an official, managerial, or professional occupation (CSCO major group 1 or 2) when the child was fourteen; less advantaged children are all others. Source: [unified/code/u14_sample_facts.py](#).

A.3 Variable definitions

Table A4: Main variables.

Variable	Definition
H_i	Equal to one if either parent held an occupation in CSCO major group 1 or 2 when the child was fourteen; retrospective reports from the 2012, 2020 and 2022 waves.
M_i	Sixteen or more completed years of schooling (a four-year college degree).
$\bar{y}_{io}, \bar{y}_{ib}$	Mean of log annual labor income, net of survey-year effects, over a child's waves at ages 30 and over (the level of earnings; at least two such waves) and over the waves at ages 22 to 29, among the 2014 to 2022 waves with positive income; incomes below 1,000 yuan and the top 1 percent within each wave are dropped.
$\overline{\text{ParentInc}}_i$	Mean of log family income of the parental household across all observed waves; a second version uses the linked parents' own labor income.
Z_i	Provincial intensity of the expansion multiplied by the exposure ratio of the child's birth cohort; the province and the birth year are those recorded in the first wave in which the child is observed at ages 22 to 55.
Professional occupation	The child's own occupation in CSCO major group 1 or 2; the outcome of the paper is the share of the child's waves at ages 30 and over in such an occupation.
Controls of the expansion design	H_i , $\overline{\text{ParentInc}}_i$, gender, urban residence, province effects, one indicator per birth year, an indicator for the row at 30 and over, and its product with H_i .

B The earnings process

B.1 Variances and first-order autocovariances by wave

Figure 1 of the paper plots the variances of residual log earnings in each wave and the autocovariances by lag for the two groups. Table A5 gives the variances and the first-order autocovariances behind it, with their bootstrap standard errors.

Table A5: Empirical autocovariances of residual log earnings, pooled and by group.

	Variances				
	2014	2016	2018	2020	2022
Pooled	0.49 (0.02)	0.58 (0.04)	0.42 (0.02)	0.46 (0.02)	0.41 (0.02)
Advantaged	0.47 (0.06)	0.49 (0.09)	0.40 (0.04)	0.44 (0.04)	0.43 (0.05)
Less advantaged	0.50 (0.03)	0.60 (0.03)	0.43 (0.02)	0.46 (0.02)	0.40 (0.02)
	First-order autocovariances				
	2014, 2016	2016, 2018	2018, 2020	2020, 2022	
Pooled	0.11 (0.04)	0.15 (0.02)	0.18 (0.01)	0.19 (0.01)	
Advantaged	0.21 (0.14)	0.15 (0.04)	0.17 (0.02)	0.24 (0.04)	
Less advantaged	0.10 (0.03)	0.15 (0.02)	0.18 (0.01)	0.18 (0.01)	

Notes: Children with at least three waves of earnings, 2014 to 2022 panel; residuals demeaned within group and year. Bootstrap standard errors in parentheses (200 replications that resample children). Across all fifteen pairwise moments, including the higher lags not displayed, no gap between the two groups exceeds two standard errors; the largest standardized gap is 1.58, at the (2014, 2022) covariance. Source: [income_dynamics/code/m5_md_weighted_and_ridge.py](#).

B.2 Joint test of moment equality

Section 3.3 of the paper reports that a joint test does not reject equality of the fifteen variances and autocovariances of the two groups. The test takes the vector of fifteen gaps between advantaged and less advantaged children in the moments of the 2014 to 2022 panel (residuals demeaned within group and year), estimates its covariance from a bootstrap that resamples children within each group, and computes the quadratic form of the gaps in the inverse of that covariance. The statistic is 17.0 on fifteen degrees of freedom ($p = 0.32$); a bootstrap-calibrated version that compares the statistic with the distribution of centered bootstrap draws gives $p = 0.311$ with 2,000 draws. Source: [income_dynamics/code/m16_joint_moment_test.py](#).

B.3 What the design could have detected

Table A6: Minimum detectable differences between the two groups.

Parameter	Gap, advantaged minus less advantaged children	Standard error	Minimum detectable difference
Permanent variance	−0.046	0.030	0.08
Variance of transitory shocks	−0.011	0.028	0.08
Transitory persistence	0.212	0.084	0.24
Permanent share	−0.092	0.068	0.19
Arellano–Bond persistence interaction	—	0.094	0.26

Notes: Standard errors of the gaps from the bootstrap of Table 3 of the paper; the last row uses the two-step sandwich standard error of the interaction in Table 4 of the paper. The minimum detectable difference at 80 percent power and the 5 percent level is 2.8 times the standard error. Source: [income_dynamics/code/m13_mde_group_null.py](#).

B.4 Alternative error structures

The paper fits a permanent effect plus an AR(1) transitory component. Table A7 reports the minimized criterion when the transitory component is instead ARMA(1,1) or MA(1), and when the permanent effect is replaced by a random walk. The richer transitory structures leave the criterion where it was, and the random walk fits worse in every sample.

Table A7: Minimized criterion under alternative error structures.

Sample	Permanent effect plus AR(1)	Permanent effect plus ARMA(1,1)	Permanent effect plus MA(1)	Random-walk permanent component
Pooled	29.3	28.9	29.8	35.3
Advantaged children	2.7	2.6	3.1	5.6
Less advantaged children	29.8	29.5	29.9	32.8

Notes: Minimum distance with weights equal to the number of children observed in both waves of each moment, on all fifteen variances and autocovariances of the 2014 to 2022 panel. Criterion values are comparable within a row. Source: [unified/code/u7_error_structures.py](#).

B.5 Diagnostics for the dynamic-panel estimate

Table A8 reports the serial-correlation tests behind Table 4 of the paper. On either panel, the test of second-order serial correlation does not reject the absence of second-order correlation on which instruments dated two waves back rely. The five-wave estimates are computed three times: in Python, in a Mata implementation that builds the same moments, and in Stata’s `xtabond2` under the same collapsed instrument layout with Windmeijer-corrected standard errors. Python and Mata agree to four decimals on both coefficients, their standard errors, and the Hansen statistic; `xtabond2` gives 0.058 (SE 0.036) and 0.010 (SE 0.096) for the two coefficients of Table 4 of the paper, and a Hansen statistic of 2.34 against 2.38. For the system GMM row, Python and `xtabond2` give persistence of 0.109 and 0.111, and the Hansen test rejects in both (20.86 and 20.49 on four degrees of freedom). The Stata files are in [income_dynamics/code/stata](#), and [income_dynamics/output/tables/stata_abgmm_parity.csv](#) collects the three sets of estimates.

Table A8: Serial-correlation tests for the Arellano–Bond estimates.

Panel	Persistence, main	Persistence, instruments		
	estimate (SE)	$m_1(z)$	$m_2(z[p])$	lagged three or more waves (SE)
Five waves, 2014–2022	0.058 (0.036)	−5.57	0.68 [0.50]	−0.81 (0.89)
Seven waves, 2010–2022 (re-constructed)	0.062 (0.033)	−7.17	−0.08 [0.93]	−0.29 (0.42)

Notes: Two-step difference GMM as in Table 4 of the paper. m_1 and m_2 are cluster-robust statistics for first- and second-order serial correlation of the differenced residuals; m_1 is negative by construction. The last column re-estimates with instruments dated three waves back and earlier; with persistence near 0.06, lagged levels three or more waves back carry little of the variation in the lagged difference, and the standard errors are correspondingly larger. Source: [income_dynamics/code/m17_ab_diagnostics.py](#).

B.6 Nonlinear persistence across runs of the estimator

The nonlinear state equation of Section 3.5 of the paper is estimated by stochastic EM, so its estimates vary with the random draws. Table A9 sets the average persistence reported in the paper beside the mean and the range over ten further runs with different seeds. In every one of the ten runs the average persistence of advantaged children is above that of less advantaged children.

Table A9: Average nonlinear persistence: the run reported in the paper and ten further runs.

Children	Run reported	Ten further runs		
	in the paper	Mean	Smallest	Largest
All children with at least three waves	0.46	0.48	0.42	0.54
Advantaged children	0.54	0.53	0.52	0.56
Less advantaged children	0.44	0.45	0.40	0.49

Notes: Persistence $\partial Q(s, u)/\partial s$ averaged uniformly over a 5×5 lattice of quantiles of the state and of the shock rank, 2014 to 2022 panel. Sources: [income_dynamics/code/ws_05b_abb_stochastic_em.py](#) (the run reported in the paper) and [unified/code/u15_nonlinear_persistence_runs.py](#) (ten runs).

C Attenuation: the schedule, rank-rank slopes, and the linked parents

Figure A1 plots the schedule of Table 6 of the paper out to ten waves. Part II derives the exact schedule, gives the closed form for the number of waves needed to reach a target, and reports bootstrap intervals.

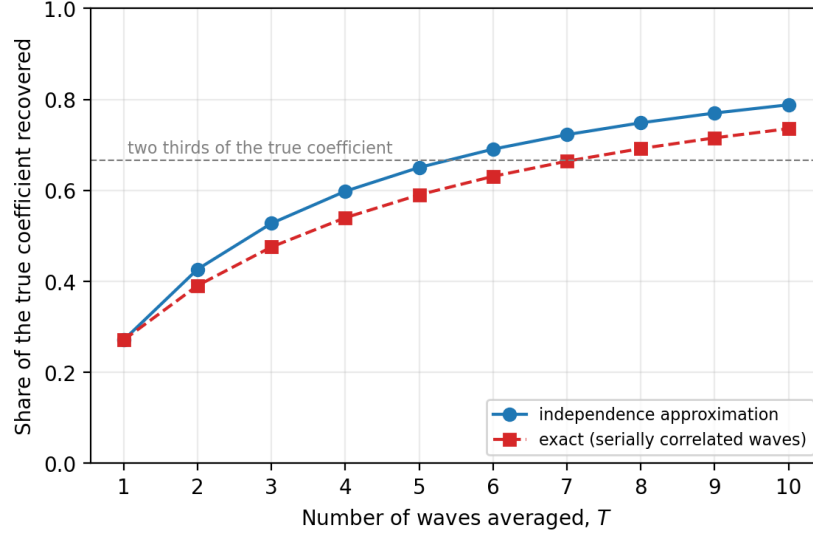


Figure A1: Share of the coefficient on permanent income recovered by an average of T waves, at the pooled estimates of Table 3 of the paper.

C.1 Rank-rank slopes

Table A10: Attenuation of rank-rank slopes by number of waves averaged (simulation from the fitted process).

Waves averaged	Share recovered, levels (exact)	Rank-rank slope, child's rank from permanent income	Rank-rank slope, child's rank from three waves
1	0.27	0.52	0.36
2	0.39	0.62	0.43
3	0.48	0.68	0.47
4	0.54	0.73	0.51
5	0.59	0.77	0.53
6	0.63	0.79	0.55
7	0.66	0.81	0.56

Notes: The rank-rank columns give the ratio of the slope under a T -wave measure of parental income to the slope under permanent income. The last column also measures the child's rank from three waves. Part II gives a closed form under normality that reproduces the third column to within 0.004. Source: [income_dynamics/code/m8_rank_attenuation_sim.py](#).

C.2 The earnings process of linked parents

Table A11: The earnings process of linked working parents.

Quantity	Value
Parents (person-waves)	542 (1,878)
Variance of residual log earnings	0.690
First-order autocovariance	0.301
Autocovariance at longer lags	0.294
Permanent share	0.43 [0.35, 0.49]
Transitory persistence	0.02

Notes: Linked parents aged sixty or younger with three or more waves of positive labor income over 2014 to 2022; log income is residualized as for the children. The permanent share is the autocovariance at longer lags relative to the variance; the interval is from a bootstrap that resamples parents. This is the parents' earnings process used in the second row of panel B of Table 6 of the paper. Source: [income_dynamics/code/m14_parent_process.py](#).

D The premium

D.1 The specifications behind Figure 3 of the paper

Table A12: The premium across fifteen specifications.

Specification	Estimate	95 percent interval	Children
Correlated random effects, family-income control (baseline)	0.088	[0.032, 0.144]	2,774
Correlated random effects, parents' own labor income as the control	0.153	[0.087, 0.219]	2,185
Correlated random effects, income deflated by provincial CPI	0.088	[0.032, 0.144]	2,774
Correlated random effects, indicator from the most frequent retrospective report	0.088	[0.032, 0.144]	2,774
Correlated random effects, indicator widened to CSCO groups 1–3	0.061	[0.008, 0.114]	2,774
Correlated random effects, indicator narrowed to CSCO group 1	0.057	[−0.021, 0.135]	2,774
Correlated random effects, region effects in place of province effects	0.081	[0.023, 0.138]	2,774
Correlated random effects, family income from the 2010 and 2012 waves only	0.107	[0.047, 0.167]	2,550
Correlated random effects, family income from waves with the child aged 40 or less	0.070	[0.008, 0.132]	2,353
Correlated random effects, the same with the child aged 35 or less	0.067	[−0.001, 0.135]	1,957
Correlated random effects, children recorded as rural in most of their waves	−0.002	[−0.122, 0.117]	1,151
Correlated random effects, children recorded as urban in most of their waves	0.113	[0.050, 0.176]	1,623
Cross-section, parental party membership added	0.070	[0.009, 0.130]	2,766
Cross-section, parental state-sector employment added	0.077	[−0.016, 0.171]	937
Entropy balancing	0.072	[0.011, 0.132]	2,774

Notes: Coefficient on the indicator for advantaged children. No specification controls for parental education. Children with a level of earnings at ages 30 and over (2,774 in the baseline). Correlated random-effects rows: child-wave observations at those ages, standard errors clustered on child. Cross-section rows: one observation per child; controls are gender, urban residence, age at first observation and its square, province effects, and parental income; heteroskedasticity-robust standard errors; the state-sector row is estimated on children with a parent working in 2012. Entropy balancing: bootstrap percentile interval. Source: [unified/code/u8_premium_specifications.py](#).

D.2 Age controls

Table A13: The premium under alternative age controls.

Age control	Coefficient on the indicator for advantaged children	SE
Linear in age (the paper’s specification)	0.088	(0.029)
Quadratic in age	0.088	(0.029)
Cubic in age	0.088	(0.028)
One indicator per year of age	0.087	(0.029)

Notes: Correlated random-effects regression of Section 5.2 of the paper with the family-income control, on the same 7,512 child-wave observations at ages 30 and over in every row; standard errors clustered on child. No row differs from the linear specification by more than 0.002. Source: [unified/code/u4_premium_age_controls.py](#).

E The expansion design

E.1 Cohort exposure

Table A14 lists the exposure ratio of each birth cohort. The national admissions series starts with the cohort born in 1964; the 38 children of the sample born before it are assigned the mean ratio of the cohorts born 1964 to 1980.

Table A14: National admissions and the exposure ratio by birth year.

Birth year	Year at age 18	Admissions (millions)	Trend projection	Exposure ratio	Birth year	Year at age 18	Admissions (millions)	Trend projection	Exposure ratio
1964	1982	0.31	0.25	1.28	1981	1999	1.55	1.08	1.44
1965	1983	0.39	0.30	1.32	1982	2000	2.21	1.13	1.96
1966	1984	0.47	0.34	1.38	1983	2001	2.68	1.18	2.28
1967	1985	0.62	0.39	1.57	1984	2002	3.20	1.22	2.62
1968	1986	0.57	0.44	1.29	1985	2003	3.82	1.27	3.00
1969	1987	0.62	0.49	1.26	1986	2004	4.47	1.32	3.38
1970	1988	0.67	0.54	1.24	1987	2005	5.05	1.37	3.68
1971	1989	0.60	0.59	1.01	1988	2006	5.46	1.42	3.84
1972	1990	0.61	0.64	0.95	1989	2007	5.66	1.47	3.85
1973	1991	0.62	0.69	0.90	1990	2008	6.08	1.52	4.00
1974	1992	0.75	0.74	1.02	1991	2009	6.39	1.57	4.08
1975	1993	0.92	0.78	1.18	1992	2010	6.62	1.62	4.09
1976	1994	0.90	0.83	1.08	1993	2011	6.82	1.67	4.09
1977	1995	0.93	0.88	1.05	1994	2012	6.89	1.71	4.02
1978	1996	0.97	0.93	1.04	1995	2013	7.00	1.76	3.97
1979	1997	1.00	0.98	1.02	1996	2014	7.21	1.81	3.98
1980	1998	1.08	1.03	1.05	1997	2015	7.38	1.86	3.97

Notes: Admissions to regular higher education in the year the cohort turned eighteen, in millions; the trend projection extends the pre-reform path; the exposure ratio is their quotient. Source: [college_expansion/data/external/cohort_exposure.csv](#).

E.2 Provincial intensity

Table A15 lists the five provincial measures. The first three are built from the predicted inflow of college graduates into the province’s workforce (Ma, 2024): the inflow divided by the 2005 stock of

college-educated workers (the main measure), the inflow divided by 2005 employment, and the log of the implied growth of the college-educated stock to 2010. The fourth is predetermined: the share of each province’s adults born 1956 to 1976 who had completed senior secondary school, measured in the 2000 census. The fifth is the placebo, the growth of entrance-examination registrations between 1997 and 1998, before the reform.

Provincial registration counts come from the annual *Chinese Educational Statistical Yearbook*, accessed through the CNKI statistical-data platform; `college_expansion/code/ws_extract_cnki_cee.py` parses the yearbook tables into a province-by-year panel. One entry is corrected: the 1997 total for Inner Mongolia in the source export disagrees with adjacent years and with the counts of men and women printed beside it, and is replaced by their sum, 66,674.

Table A15: Provincial intensity measures.

Province	Inflow per 2005 college-educated worker	Inflow per 2005 worker	Log growth of the college-educated stock to 2010	Share with senior secondary schooling (2000 census)	Registration growth 1998/1997 (placebo)
Shanghai	0.972	0.209	0.679	0.450	0.137
Tianjin	0.954	0.149	0.670	0.417	0.194
Tibet	0.876	0.009	0.629	0.089	0.095
Beijing	0.746	0.238	0.557	0.524	0.292
Jilin	0.731	0.065	0.549	0.291	0.113
Hubei	0.707	0.044	0.535	0.252	0.217
Shaanxi	0.683	0.054	0.521	0.244	0.080
Liaoning	0.676	0.071	0.516	0.258	0.171
Sichuan	0.576	0.029	0.455	0.163	-0.153
Jiangxi	0.541	0.031	0.433	0.191	0.087
Anhui	0.534	0.028	0.428	0.140	0.139
Hunan	0.505	0.029	0.409	0.207	0.055
Heilongjiang	0.504	0.050	0.408	0.276	0.080
Zhejiang	0.476	0.031	0.389	0.182	0.126
Jiangsu	0.469	0.036	0.385	0.241	0.143
Shanxi	0.468	0.039	0.384	0.240	0.090
Guizhou	0.438	0.020	0.363	0.127	0.041
Yunnan	0.410	0.016	0.344	0.131	0.080
Fujian	0.407	0.028	0.341	0.186	0.091
Gansu	0.406	0.023	0.341	0.196	0.034
Shandong	0.382	0.020	0.324	0.205	0.112
Hebei	0.372	0.023	0.316	0.197	0.210
Guangxi	0.370	0.021	0.315	0.200	0.067
Henan	0.348	0.019	0.298	0.192	0.072
Inner Mongolia	0.313	0.033	0.272	0.265	0.082
Qinghai	0.312	0.032	0.271	0.227	0.057
Guangdong	0.280	0.021	0.247	0.248	0.181
Ningxia	0.266	0.029	0.236	0.247	0.093
Xinjiang	0.241	0.030	0.216	0.290	0.179

Notes: The 29 provinces of the sample, ordered by the main measure (first column). Sources: `college_expansion/data/external/expansion_intensity_real.csv`, `college_expansion/data/external/census_hs_stock.csv`, `college_expansion/data/external/expansion_intensity_canonical.csv`.

E.3 Definitions of the occupation outcome

Section 6.5 of the paper measures the child’s professional occupation as the share of the child’s waves at ages 30 and over in which the reported occupation is professional. Table A16 compares that definition with four others, each measured on the same waves, in the regression on the rows at 30 and over only (equation (11) of the paper). The reduced form is positive under every definition, and it is smallest for an occupation held in any single wave, the noisiest measure of a lasting position.

Table A16: The occupation reduced form under five definitions of the outcome.

Definition of the outcome	Mean	Reduced form (SE)	Wild-cluster p
Ever observed in a professional occupation	0.29	0.090 (0.029)	0.025
In a professional occupation in most observed waves	0.15	0.099 (0.028)	0.011
Share of observed waves in a professional occupation	0.19	0.098 (0.024)	0.017
In a professional occupation at the last observed wave	0.19	0.123 (0.028)	0.012
In a professional occupation in two or more waves	0.16	0.108 (0.029)	0.018

Notes: The 2,774 children with a level of earnings at ages 30 and over, one observation per child, specification of equation (11) of the paper. Province-clustered standard errors; wild cluster bootstrap p -values with 9,999 replications. Source: [unified/code/u3_online_appendix.py](#).

E.4 Trends before the reform in the census

Every birth cohort from 1956 to 1976 completed its college window before the reform. Table A17 compares the cohort slope of university completion between provinces in the top and bottom thirds of intensity, in the 1990 and 2000 census samples. Over the twenty years between the 1956 and 1976 cohorts the difference cumulates to under one percentage point.

Table A17: Difference in pre-reform cohort slopes of university completion, provinces in the top third of intensity against the bottom third.

Census	Difference in cohort slope	SE	p	Province-cohort cells
2000	0.00043	0.00024	0.072	420
1990	0.00030	0.00023	0.196	420

Notes: One percent census samples, ages 24 to 44, produced by the National Bureau of Statistics of China and accessed through IPUMS International (Ruggles et al., 2025). Slopes are in completion rate per birth year; standard errors clustered on province. Source: [college_expansion/code/p16_census_pretrend.py](#).

E.5 The least squares return to college by age

Section 2.2 of the paper measures the level of earnings at ages 30 and over because the return to college is not yet visible in the early twenties. Table A18 reports the least squares return to college by age on all child-wave observations of the sample, in which children are observed from age 22.

Table A18: The least squares return to college by age at observation.

Age at observation	Least squares return to college	SE	Child-wave observations
22–25	−0.021	(0.047)	1,815
26–29	0.369	(0.035)	2,702
30–34	0.465	(0.033)	2,867
35–55	0.414	(0.035)	5,291

Notes: Child-wave observations of the sample (ages 22 to 55, at least two waves). Controls: family background, parental income, gender, urban residence, province effects, survey-year effects, one indicator per year of age. Standard errors clustered on child. Source: [unified/code/u14_sample_facts.py](#).

E.6 Earnings at ages 30 and over: variants of the specification and checks

Section 6.4 of the paper estimates the return to college in a regression with two rows per child, the mean of earnings at ages 30 and over and the mean below 30, and reports the coefficient of the row at 30 and over (equation (12) of the paper). Table A19 sets that specification beside its variants: the regression on the rows at 30 and over only (equation (11)), the two-row regression in which a row below 30 is kept only if it averages two or more waves, and the same regressions with each row weighted by the precision that the earnings process of Section 3 of the paper implies for a mean of that many waves (equation (14)). The return lies between 0.63 and 0.70 in every row, and the wild-cluster p -value of the reduced form is below 0.005.

Table A20 reports the checks of the earnings result, with unweighted and with weighted rows: neither the earnings nor the occupations of children with nine or fewer, or with twelve or fewer, years of schooling respond to the instrument, and trade controls, cohort trends specific to each of three regions, the assignment of the instrument by the province at age twelve, and cohort exposure taken at age seventeen or nineteen instead of eighteen leave the reduced form in place. The last row leaves out the cohorts born 1973 to 1976, the pre-reform cohorts whose college completion has the largest gradient on provincial intensity in the event study of Figure 5 of the paper. With a linear cohort trend for each province, the first stage has $F = 16.2$, the earnings reduced form is 0.100 (SE 0.065), and the return to college is 0.77. Among the cohorts born before the reform, the interaction of provincial intensity with birth year has a wild-cluster p -value of 0.56 for earnings and 0.75 for college completion. Leaving out one province at a time keeps the return within [0.65, 1.11]. Table A21 repeats the estimates by family background of Table 11 of the paper with weighted rows.

The completion indicator counts sixteen or more years of schooling. In the same two-row regression, the coefficient on the instrument is -0.060 (SE 0.030, wild-cluster $p = 0.29$) when the outcome is exactly fifteen years of schooling, the three-year degree, 0.127 (SE 0.035) when it is fifteen or more years, and 0.186 (SE 0.044) for the completion indicator of the paper ([unified/output/u9_G_three_year_completion.csv](#)).

Table A19: The instrumented return to college at ages 30 and over: the main specification and its variants.

Specification	Children	Rows at 30 and over	First-stage F	Earnings reduced form (SE) [wild p]	Return to college
Two rows per child, unweighted (the paper's main specification)	4,536	2,774	18.0	0.128 (0.018) [0.002]	0.68
Rows at 30 and over only	2,774	2,774	21.2	0.136 (0.018) [0.003]	0.70
Two rows per child, the row below 30 kept only if it averages two or more waves	4,187	2,774	19.6	0.123 (0.020) [0.002]	0.64
Two rows per child, weighted by the precision implied by the earnings process	4,536	2,774	16.9	0.118 (0.019) [0.003]	0.63
Rows at 30 and over only, weighted in the same way	2,774	2,774	19.4	0.125 (0.020) [0.004]	0.64

Notes: The sample of the paper (4,536 children observed in at least two waves at ages 22 to 55), main intensity measure. Every entry is the estimate for the row at 30 and over. The weight of a row that averages n waves is the inverse of $\hat{\sigma}_\eta^2 + \hat{\sigma}_\varepsilon^2 / ((1 - \hat{\rho}_v^2) n)$ at the pooled estimates of Table 3 of the paper. Province-clustered standard errors; wild cluster bootstrap p -values (9,999 replications). Source: [unified/code/u9_mature_age_earnings.py](#).

Table A20: Checks of the earnings result at ages 30 and over.

Sample or added control	Children	Unweighted rows		Rows weighted by precision	
		Reduced form (SE) [wild p]	Return to college	Reduced form (SE) [wild p]	Return to college
Children with nine or fewer years of schooling	1,811	0.006 (0.096) [0.959]	—	−0.028 (0.097) [0.808]	—
Children with twelve or fewer years of schooling	2,688	0.068 (0.055) [0.272]	—	0.042 (0.058) [0.509]	—
Children with nine or fewer years of schooling, professional occupation as the outcome	1,809	0.017 (0.019) [0.422]	—	0.027 (0.020) [0.250]	—
Children with twelve or fewer years of schooling, professional occupation as the outcome	2,684	0.016 (0.020) [0.465]	—	0.018 (0.021) [0.408]	—
Coastal province $\times$ exposure added	4,536	0.131 (0.022) [0.003]	0.71	0.121 (0.023) [0.006]	0.65
Export share $\times$ exposure added	4,536	0.127 (0.020) [0.002]	0.68	0.117 (0.018) [0.001]	0.62
Both added	4,536	0.140 (0.025) [0.007]	0.75	0.129 (0.022) [0.006]	0.69
Linear cohort trends for each of three regions added	4,536	0.133 (0.025) [0.003]	0.72	0.123 (0.026) [0.007]	0.67
Linear cohort trend for each province added	4,536	0.100 (0.065) [0.365]	0.77	0.069 (0.068) [0.542]	0.55
Instrument assigned by the province at age twelve	4,536	0.149 (0.022) [0.003]	0.74	0.137 (0.023) [0.005]	0.68
Cohort exposure taken at age seventeen instead of eighteen	4,536	0.149 (0.022) [0.001]	0.74	0.139 (0.023) [0.001]	0.68
Cohort exposure taken at age nineteen instead of eighteen	4,536	0.108 (0.024) [0.024]	0.64	0.099 (0.022) [0.026]	0.58
Cohorts born 1973 to 1976 left out	4,192	0.127 (0.020) [0.002]	0.60	0.118 (0.019) [0.002]	0.56

Notes: Two-row specification of Table 10 of the paper, main intensity measure; every entry is the estimate for the row at 30 and over. The first four rows restrict the regression to children with the stated schooling; the outcome is earnings in the first two and, in the next two, the share of the child's waves at ages 30 and over in a professional occupation. Province-clustered standard errors; wild cluster bootstrap p -values (9,999 replications). The return to college is the two-stage least squares estimate of the same regression. Source: [unified/code/u9_mature_age_earnings.py](#).

Table A21: Returns to college for all children and by family background, with unweighted and with weighted rows.

Family background	First-stage	Earnings reduced form	Return to	Anderson–Rubin set	
	F	(SE) [wild p]	college (SE)	95 percent	90 percent
<i>Unweighted rows</i>					
All children	18.0	0.128 (0.018) [0.002]	0.68	[0.50, 4.20]	[0.55, 2.30]
Less advantaged children	16.9	0.115 (0.018) [0.004]	0.62 (0.19)	[0.45, 4.45]	[0.45, 2.30]
Advantaged children	12.3	0.191 (0.058) [0.042]	1.14 (0.36)	$(-\infty, -0.50] \cup [0.25, \infty)$	$(-\infty, -3.20] \cup [0.60, \infty)$
<i>Rows weighted by precision</i>					
All children	16.9	0.118 (0.019) [0.003]	0.63	[0.50, 4.70]	[0.50, 2.15]
Less advantaged children	15.3	0.105 (0.017) [0.005]	0.57 (0.17)	[0.40, 5.00]	[0.45, 2.20]
Advantaged children	13.6	0.184 (0.059) [0.063]	1.07 (0.32)	unbounded	$(-\infty, -2.60] \cup [0.45, \infty)$

Notes: Specification of Table 11 of the paper; every entry is the estimate for the row at 30 and over. The rows for all children come from the regression of Table 10 of the paper under the main intensity measure; for all children the inference on the return is the Anderson–Rubin set, and no standard error is printed. Anderson–Rubin sets over a grid from -6 to 6 with step 0.05 ; a set is unbounded when the wild-cluster test does not reject, at that level, a zero coefficient on the group’s instrument in the first stage for the group’s own college completion, and a set made of two rays is printed as their union. Sources: [unified/code/u9_mature_age_earnings.py](#) (all children) and [unified/code/u12_returns_by_family_background.py](#).

E.7 Inference

Intensity varies by province, so inference rests on 29 clusters. The p -values of the reduced forms are from a wild cluster bootstrap with the null imposed: the outcome is regressed on the controls alone, the residuals are multiplied by one Rademacher draw per province, and the cluster-robust t -statistic on the instrument is recomputed in each of 9,999 replications. In the two-row regression the instrument tested is that of the row at 30 and over, and the instrument of the row below 30 is among the controls. The Anderson–Rubin set for the return to college collects the values b_0 at which the same test does not reject a zero coefficient on that instrument in the regression of earnings minus b_0 times college completion at 30 and over, over a grid from -6 to 6 with step 0.05 . As b_0 grows without bound the test becomes the test of a zero first stage, so a set is unbounded on a side when the test of a zero first stage does not reject at that level. The bootstrap weights are drawn once and reused at every point of the grid. The two-row regression has two endogenous regressors, college completion in each row, and two instruments, so two-stage least squares and LIML coincide. In the regression on the rows at 30 and over only, with one instrument and one endogenous regressor, the cluster-robust first-stage F equals the effective F of [Montiel Olea and Pflueger \(2013\)](#).

Table A22 reports randomization inference, which does not rest on the number of provinces being large. The 29 provincial values of the main intensity measure are reassigned at random among the provinces 4,999 times; the cohort exposure, the outcomes, and the controls of every child stay as they are. For each reassignment the two-row regression is estimated again, and the p -value is the share of reassignments, the actual assignment counted among them, whose province-clustered t -statistic on the instrument at 30 and over is at least as large in absolute value as the actual one. A second scheme reassigns intensity only among the provinces of the same region (ten provinces in the east, eight in the center, and eleven in the west). Inverting the within-region test over the grid of the Anderson–Rubin sets gives a set for the return to college; the test at each

value assumes the same return for every child.

Table A22: Randomization inference for the earnings reduced form, the first stage, and the return to college.

Reassignment of provincial intensity	Earnings reduced form		First stage		Set for the return to college	
	As large as actual	p	As large as actual	p	95 percent	90 percent
Among all 29 provinces	0	0.0002	2	0.0006	—	—
Within region (east, center, west)	0	0.0002	2	0.0006	[0.45, 1.30]	[0.50, 1.10]

Notes: Two-row specification of the paper on all 4,536 children, main intensity measure; 4,999 reassignments under each scheme. Reached: the number of reassignments whose t -statistic is at least as large in absolute value as the actual one; p is that number plus one, divided by 5,000. The set collects the values of the return, on a grid from -6 to 6 with step 0.05 , that the within-region test does not reject. Source: block R of [unified/code/u9_mature_age_earnings.py](#).

The estimates by family background (Table 11 of the paper) come from one regression in which the instrument enters once for each group and row, province effects are common to the two groups, birth-year effects are specific to each group, and the indicator for the row at 30 and over and its product with family background are among the controls. The return of each group is estimated by two-stage least squares with four endogenous regressors, the college completion of each group in each row, and the four instruments. The Anderson–Rubin set of a group collects the values b_0 at which the wild-cluster test does not reject a zero coefficient on that group’s instrument at 30 and over, over a grid from -6 to 6 with step 0.05 . As b_0 grows without bound this test becomes the test of a zero coefficient on the group’s instrument in the first stage for the group’s own college completion, so the set of a group is unbounded when that test does not reject at that level. The first stages printed in Table 11 of the paper are the coefficients on each group’s instrument in the first stage whose dependent variable is the college completion of all children. Source: [unified/code/u12_returns_by_family_background.py](#).

F The decomposition

F.1 Across estimates of the premium

Table 12 of the paper evaluates the identity at the entropy-balancing estimate of the premium, for which the completion gap is measured under the same weights. Table A23 gives the college share at each of the three estimates of the premium in Table 7 of the paper, at the least squares return and at the instrumented return for all children. The share is smallest at the regression estimate that controls for the parents’ labor income, the larger of the two regression estimates of the premium: 36 percent at the least squares return and 55 percent at the instrumented return.

Table A23: The college share of the premium across estimates of the premium.

Estimate of the premium	Premium	College share at the return to college		Return at which college explains	
		Least squares	Instrumented	Half	All
Correlated random effects, family income	0.088	0.63	0.95	0.358	0.717
Correlated random effects, parents' labor income	0.153	0.36	0.55	0.626	1.252
Entropy balancing	0.072	0.76	1.16	0.294	0.588

Notes: Level of earnings at ages 30 and over. College share is the return to college times the gap in college completion of 0.122, divided by the premium. A share above one means the college channel exceeds the premium. Source: [unified/code/u3_online_appendix.py](#).

F.2 Inside the entropy-balanced comparison

Inside the comparison of advantaged children with reweighted less advantaged children, every term of the identity is a directly measured mean, and the premium of 0.072 splits exactly into three parts (Section 7 of the paper). Graduates out-earn children without a degree by 0.541 log points among advantaged children and by 0.517 among reweighted less advantaged children; 33.55 percent of the former and 21.30 percent of the latter hold a degree. The additional degrees of advantaged children, valued at the return among advantaged graduates, account for 0.066; the degrees held in both groups, worth 0.024 more to advantaged children, account for 0.005; and the earnings gap between the two groups among children without a degree is 0.001. Source: [unified/code/u16_identity_in_balanced_comparison.py](#).

Part II. Formal results

This part states and proves the results that Sections 3, 4, and 7 of the paper use without proof. Section G derives the attenuation schedule of Section 4 of the paper from the estimated earnings process: the exact adjustment for serial correlation, the number of waves needed to reach a target share of the coefficient, and a closed form for the simulated rank-rank factors. Section H treats the minimum-distance estimator of Section 3.3 of the paper: identification, the ridge of Figure 2 of the paper, the distribution of the criterion under fixed weights, and a decomposition of the criterion into a stationarity component and a shape component. Section I writes the accounting identity of Section 7 of the paper in potential outcomes and shows what its remainder contains. Every number that is not copied from the paper is produced by [formal_results/code/appendix_theory_computations.py](#) in the repository, and [online_appendix/build/check_part2_numbers.py](#) asserts each printed number against its source.

G The attenuation schedule

G.1 Setup

Residual log earnings of child i in wave t follow equation (4) of the paper,

$$\tilde{y}_{it} = \eta_i + v_{it}, \quad v_{it} = \rho v_{i,t-1} + \varepsilon_{it}, \quad |\rho| < 1, \quad (\text{G.1})$$

with η_i a permanent component, ε_{it} serially uncorrelated with mean zero and variance σ_ε^2 , and $\{\varepsilon_{it}\}$ independent of η_i . The persistence parameter is written ρ_v in the paper; this part writes ρ . Under stationarity the transitory component has variance $\sigma_v^2 = \sigma_\varepsilon^2/(1 - \rho^2)$ and autocovariance function $\gamma_k = \rho^k \sigma_v^2$, $k \geq 0$, so the autocovariances of $\tilde{y}$ are those of equation (5) of the paper,

$$\omega_k \equiv \text{Cov}(\tilde{y}_{it}, \tilde{y}_{i,t+k}) = \sigma_\eta^2 + \rho^k \sigma_v^2. \quad (\text{G.2})$$

One period is one CFPS wave (two years). The permanent share is $s = \sigma_\eta^2/(\sigma_\eta^2 + \sigma_v^2)$. Table 3 of the paper estimates $(\hat{\sigma}_\eta^2, \hat{\sigma}_\varepsilon^2, \hat{\rho}) = (0.125, 0.326, 0.16)$ on the pooled sample, so $\hat{\sigma}_v^2 = 0.335$ and $\hat{s} = 0.27$.

G.2 The variance of a wave average

Proposition G.1 (Variance of a T -wave mean of a stationary AR(1)). *Let $\bar{v}_{iT} = T^{-1} \sum_{t=1}^T v_{it}$. Then*

$$\text{Var}(\bar{v}_{iT}) = \frac{\sigma_v^2}{T} f_T(\rho), \quad f_T(\rho) = 1 + 2 \sum_{k=1}^{T-1} \left(1 - \frac{k}{T}\right) \rho^k = \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^T)}{T(1 - \rho)^2}. \quad (\text{G.3})$$

For $\rho \in (0, 1)$: $f_1 = 1$; f_T is strictly increasing in T with limit $(1 + \rho)/(1 - \rho)$; f_T/T is strictly decreasing in T ; and for $T \geq 2$, f_T is strictly increasing in ρ .

Proof. $\text{Var}(\bar{v}_{iT}) = T^{-2} \sum_{s=1}^T \sum_{t=1}^T \gamma_{|s-t|} = T^{-2} [T\gamma_0 + 2 \sum_{k=1}^{T-1} (T - k)\gamma_k]$, and substituting $\gamma_k = \rho^k \sigma_v^2$ gives the sum form of f_T . For the closed form, use $\sum_{k=1}^{T-1} \rho^k = \rho(1 - \rho^{T-1})/(1 - \rho)$ and $\sum_{k=1}^{T-1} k\rho^k = \rho(1 - T\rho^{T-1} + (T-1)\rho^T)/(1 - \rho)^2$; collecting terms,

$$f_T = 1 + \frac{2\rho(1 - \rho^{T-1})}{1 - \rho} - \frac{2\rho(1 - T\rho^{T-1} + (T-1)\rho^T)}{T(1 - \rho)^2},$$

and the two fractions combine to $\frac{2\rho}{1 - \rho} - \frac{2\rho(1 - \rho^T)}{T(1 - \rho)^2}$, which with the leading 1 is (G.3). In the sum form each summand $(1 - k/T)\rho^k$ is positive and increasing in T and in ρ , and the number of summands grows with T , which gives the two monotonicity statements for f_T ; the limit follows from the closed form. For f_T/T , write the mean of $T + 1$ consecutive terms as the average of its $T + 1$ leave-one-out means. Each leave-one-out mean averages T terms whose pairwise distances in time are at least those of T consecutive terms, so with $\rho \in (0, 1)$ its variance is at most $\text{Var}(\bar{v}_{iT})$. The variance of an average of random variables is at most the largest of their variances,

strictly so unless they are perfectly correlated, which the leave-one-out means are not. Hence $\text{Var}(\bar{v}_{i,T+1}) < \text{Var}(\bar{v}_{iT})$. $\square$

At the pooled estimate $\hat{\rho} = 0.16$, $f_2 = 1.16$, $f_5 = 1.29$, $f_9 = 1.33$, and $f_\infty = 1.38$. Because transitory components of consecutive waves are correlated, averaging removes their variance more slowly than the independence calculation implies; the variance of a long average is up to 38 percent larger.

G.3 Attenuation

Assumption G.1 (Classical measurement structure). The outcome satisfies $c_i = a + by_i^* + e_i$, where y_i^* is permanent income with variance σ_η^2 and b is the coefficient of the linear projection of c_i on y_i^* , so that $\text{Cov}(y_i^*, e_i) = 0$. The researcher observes $x_{iT} = y_i^* + \bar{v}_{iT}$, with $\bar{v}_{iT}$ as in Proposition G.1, and regresses c_i on x_{iT} by least squares on an i.i.d. sample. The transitory components are uncorrelated with y_i^* and with e_i .

Proposition G.2 (The attenuation schedule). *Under (G.1) and Assumption G.1, the least squares slope $\hat{b}_T$ satisfies*

$$\text{plim } \hat{b}_T = b \lambda_T, \quad \lambda_T = \frac{\sigma_\eta^2}{\sigma_\eta^2 + \sigma_v^2 f_T(\rho)/T}. \quad (\text{G.4})$$

The independence approximation of equation (8) of the paper is $\lambda_T^\circ = \sigma_\eta^2/(\sigma_\eta^2 + \sigma_v^2/T)$, that is, (G.4) with f_T replaced by one. For $\rho \in (0, 1)$ and given $(\sigma_\eta^2, \sigma_v^2)$: (i) $\lambda_T \leq \lambda_T^\circ$, with equality only at $T = 1$; (ii) λ_T is strictly increasing in T with limit one; (iii) $1 - \lambda_T = O(1/T)$; and (iv) for $T \geq 2$, λ_T is strictly decreasing in ρ .

Proof. The least squares slope converges to $\text{Cov}(c, x_T)/\text{Var}(x_T) = b \sigma_\eta^2/(\sigma_\eta^2 + \text{Var}(\bar{v}_T))$ because $\text{Cov}(y^*, e) = \text{Cov}(y^*, \bar{v}_T) = \text{Cov}(e, \bar{v}_T) = 0$; substitute Proposition G.1. (i) follows from $f_T \geq 1$ with equality only at $T = 1$. (ii) and (iii) follow from f_T/T decreasing in T and bounded above by f_∞/T . (iv) follows from f_T increasing in ρ . $\square$

Table A24 attaches sampling uncertainty to the schedule of Table 6 of the paper. The single-wave factor is 0.27 with a 95 percent interval of [0.23, 0.31]: even at the top of the interval a single wave recovers less than a third of the coefficient on permanent income. At seven waves the exact factor is 0.66 with interval [0.59, 0.72]. The paper's statement that six biennial waves reach two thirds uses the independence approximation; under the exact formula the crossing is at eight waves, as the footnote to equation (8) of the paper states.

G.4 Which side of the regression

Suppose the outcome is also observed as a T_c -wave average, $c_{iT_c} = c_i + \bar{w}_{iT_c}$, where $\bar{w}_{iT_c}$ is uncorrelated with y_i^* , $\bar{v}_{iT}$, and e_i . The slope of c_{iT_c} on x_{iT} then has probability limit $b \lambda_T$ regardless of T_c : noise in the dependent variable adds to the regression disturbance and is uncorrelated with the regressor, which leaves the probability limit of the slope unchanged, whereas noise

Table A24: The attenuation schedule with bootstrap intervals.

Waves T	Independence approximation λ_T°		Exact λ_T		$1/\lambda_T$
	Point	95 percent interval	Point	95 percent interval	
1	0.271	[0.23, 0.31]	0.271	[0.23, 0.31]	3.69
2	0.427	[0.37, 0.47]	0.391	[0.33, 0.45]	2.56
3	0.528	[0.47, 0.57]	0.476	[0.41, 0.54]	2.10
4	0.598	[0.54, 0.64]	0.540	[0.47, 0.60]	1.85
5	0.651	[0.60, 0.69]	0.590	[0.52, 0.65]	1.69
6	0.691	[0.64, 0.73]	0.631	[0.56, 0.69]	1.58
7	0.723	[0.68, 0.76]	0.664	[0.59, 0.72]	1.51
9	0.770	[0.73, 0.80]	0.716	[0.65, 0.77]	1.40

Notes: Equation (G.4) at the pooled minimum-distance estimate of Table 3 of the paper. Intervals are percentile intervals from 1,000 bootstrap replications that resample children, re-demean the panel, recompute the fifteen pairwise autocovariances, and re-solve the minimum-distance fit in every draw, then evaluate (G.4) at each replication's parameters. The intervals for the exact factor are wider because $f_T(\rho)$ adds the sampling variation of $\hat{\rho}$ to that of the two variances. The last column is the factor by which a researcher would multiply an estimated coefficient to undo attenuation under the exact formula, were the parameters known.

in the regressor enters the denominator as in Proposition G.2. If the roles are reversed and x_{iT} is regressed on c_{iT_c} , the slope is the coefficient of the projection of y_i^* on c_i multiplied by $\text{Var}(c_i)/\text{Var}(c_{iT_c})$, the reliability of the averaged outcome, and λ_T does not enter.

Section 4 of the paper accordingly takes the relevant T in an intergenerational elasticity regression to be the number of parental income observations, and panel B of Table 6 of the paper recomputes the correction at the earnings process of linked parents, whose permanent share is 0.43 against the children's 0.27 (Table A11).

G.5 Waves needed to reach a target

Proposition G.3 (Waves required). *For a target share $\kappa \in (0, 1)$ of the coefficient define $T^*(\kappa) = \min\{T : \lambda_T \geq \kappa\}$ and $T^{\circ*}(\kappa) = \min\{T : \lambda_T^\circ \geq \kappa\}$. Then*

$$T^{\circ*}(\kappa) = \left\lceil \frac{\kappa}{1-\kappa} \cdot \frac{\sigma_v^2}{\sigma_\eta^2} \right\rceil = \left\lceil \frac{\kappa}{1-\kappa} \cdot \frac{1-s}{s} \right\rceil, \quad (\text{G.5})$$

and for $\rho \in (0, 1)$, $T^*(\kappa) \geq T^{\circ*}(\kappa)$, with $T^*(\kappa)$ the smallest T such that $f_T(\rho)/T \leq (1 - \kappa) \sigma_\eta^2 / (\kappa \sigma_v^2)$.

Proof. $\lambda_T^\circ \geq \kappa$ is equivalent to $\sigma_\eta^2(1 - \kappa) \geq \kappa \sigma_v^2/T$, that is $T \geq \kappa \sigma_v^2 / ((1 - \kappa) \sigma_\eta^2)$; the second expression substitutes $\sigma_v^2 / \sigma_\eta^2 = (1 - s)/s$. The exact statement rearranges (G.4) in the same way with f_T/T in place of $1/T$, and $f_T \geq 1$ gives the inequality between the two thresholds. $\square$

At the pooled estimates and $\kappa = 2/3$, the threshold in (G.5) is $2 \times 0.335/0.125 = 5.36$, so $T^{\circ*} = 6$, the number of waves in Section 4 of the paper. The exact value is $T^* = 8$, because $f_8/8 = 0.166$ is the first value of f_T/T at or below 0.186. Carrying the bootstrap draws through the two thresholds gives 95 percent intervals of $[5, 7]$ waves under the approximation and $[6, 10]$

under the exact formula. Six biennial waves span a decade of panel coverage and eight span fourteen years, so the two-thirds benchmark that [Mazumder \(2005\)](#) uses for the United States is reached in this population after a decade or more of biennial coverage, as the paper says.

G.6 Rank-rank slopes

Section 4 of the paper reports by simulation that a single-wave design recovers 0.52 of the true rank-rank slope, roughly $\sqrt{\lambda_1}$, and 0.36 once the child's rank is itself measured from three waves. Table A10 gives the simulated factors for one to seven waves. The following result gives the closed form behind the simulation and explains the square root.

Proposition G.4 (Rank-rank attenuation under normality). *Let permanent parental income α and the child outcome c be jointly normal with correlation $r \in (0, 1]$, and let the measured parental income be $m_T = \alpha + \bar{v}_T$ with $\bar{v}_T$ normal, independent of (α, c) , and reliability $\text{Var}(\alpha)/\text{Var}(m_T) = \lambda_T$. Then the population rank-rank slope of c on m_T , relative to the slope of c on α , is*

$$R_T = \frac{\arcsin(r\sqrt{\lambda_T}/2)}{\arcsin(r/2)}, \quad \text{and} \quad r \rightarrow 0 \Rightarrow R_T \rightarrow \sqrt{\lambda_T}, \quad (\text{G.6})$$

with $R_T \leq \sqrt{\lambda_T}$ for every $r \in (0, 1]$. If the child outcome is itself measured as a T_c -wave average $c_{T_c} = c + \bar{w}_{T_c}$, with $\bar{w}_{T_c}$ normal, independent of $(\alpha, c, \bar{v}_T)$, and reliability $\text{Var}(c)/\text{Var}(c_{T_c}) = \lambda_{T_c}$, the factor becomes

$$\frac{\arcsin(r\sqrt{\lambda_T \lambda_{T_c}}/2)}{\arcsin(r/2)},$$

which tends to $\sqrt{\lambda_T \lambda_{T_c}}$ as $r \rightarrow 0$.

Proof. Spearman's rank correlation for a bivariate normal pair with correlation ϱ is $(6/\pi) \arcsin(\varrho/2)$ ([Pearson, 1907](#); [Kruskal, 1958](#)). Population ranks are uniform on $[0, 1]$ on both sides, so the rank-rank regression slope equals the rank correlation. The pair (m_T, c) is bivariate normal, and since

$$\text{Cov}(m_T, c) = \text{Cov}(\alpha, c) \quad \text{and} \quad \text{Var}(m_T) = \text{Var}(\alpha)/\lambda_T,$$

its correlation is $\text{Corr}(m_T, c) = r\sqrt{\lambda_T}$; taking the ratio of the two Spearman correlations gives (G.6). As $r \rightarrow 0$, $\arcsin(x) \sim x$ gives the square root. The inequality follows because $\arcsin(x)/x$ is increasing on $(0, 1]$, so

$$\frac{\arcsin(r\sqrt{\lambda_T}/2)}{r\sqrt{\lambda_T}/2} \leq \frac{\arcsin(r/2)}{r/2},$$

which rearranges to $R_T \leq \sqrt{\lambda_T}$. The two-sided statement replaces the correlation $r\sqrt{\lambda_T}$ with

$$\text{Corr}(m_T, c_{T_c}) = r\sqrt{\lambda_T}\sqrt{\lambda_{T_c}}.$$

□

Table A25: Rank-rank attenuation: closed form against simulation.

Waves of parental income T	1	2	3	4	5	6	7
λ_T (exact)	0.271	0.391	0.476	0.540	0.590	0.631	0.664
$\sqrt{\lambda_T}$	0.521	0.625	0.690	0.735	0.768	0.794	0.815
Closed form (G.6), $r = 0.30$	0.519	0.624	0.688	0.734	0.767	0.793	0.814
Simulation	0.517	0.620	0.685	0.734	0.770	0.794	0.814
Closed form, child measured with 3 waves	0.358	0.430	0.474	0.505	0.529	0.546	0.561
Simulation, child measured with 3 waves	0.358	0.429	0.473	0.506	0.532	0.549	0.561

Notes: The simulation rows are the factors of Online Appendix Table A10, printed to three decimals (400,000 Gaussian draws, $r = 0.30$). The closed-form rows evaluate (G.6) at the λ_T of the first row, which is the value the simulation uses. That value is computed from the unrounded pooled estimates and is the exact column of Table A24. The largest discrepancy between closed form and simulation is 0.004.

Table A25 shows that the closed form reproduces the simulated factors to within 0.004. Under normality the square-root rule holds up to a correction, $R_T \leq \sqrt{\lambda_T}$, of at most 0.002 at $r = 0.30$. Two consequences follow without further simulation. First, ranks attenuate less than levels, since $\sqrt{\lambda_T} > \lambda_T$, but a single-wave rank design at $\lambda_1 = 0.27$ still recovers only half the slope. Second, the factor with both sides measured with error is a product, so transitory variation in the child's income lowers rank slopes as much as transitory variation in the parents' income does, whereas Section G.4 shows that it leaves slopes in levels unchanged.

G.7 Delta-method standard errors

The bootstrap in Table A24 can be replaced by a first-order expansion when only the parameter covariance is available. Writing $\theta = (\sigma_\eta^2, \sigma_\varepsilon^2, \rho)$, $D_T = \sigma_\eta^2 + \sigma_\varepsilon^2 f_T/T$, and $g_T(\rho) = f_T(\rho)/(1 - \rho^2)$, so that $\lambda_T = \sigma_\eta^2/(\sigma_\eta^2 + \sigma_\varepsilon^2 g_T/T)$,

$$\frac{\partial \lambda_T}{\partial \sigma_\eta^2} = \frac{\sigma_\varepsilon^2 g_T/T}{D_T^2}, \quad \frac{\partial \lambda_T}{\partial \sigma_\varepsilon^2} = -\frac{\sigma_\eta^2 g_T/T}{D_T^2}, \quad \frac{\partial \lambda_T}{\partial \rho} = -\frac{\sigma_\eta^2 \sigma_\varepsilon^2 g'_T(\rho)/T}{D_T^2}, \quad (\text{G.7})$$

with $g'_T(\rho) = [f'_T(\rho)(1 - \rho^2) + 2\rho f_T(\rho)]/(1 - \rho^2)^2$ and $f'_T(\rho) = 2 \sum_{k=1}^{T-1} k(1 - k/T)\rho^{k-1}$. The delta-method variance is $\nabla \lambda'_T \hat{V}_\theta \nabla \lambda_T$. The minimum-distance estimator imposes $\sigma_\eta^2 \geq 0$, and the estimate for advantaged children lies close to that boundary (Table 3 and Figure 2 of the paper). Near that boundary neither the expansion nor the percentile bootstrap applies (Remark H.5), so Table A24 reports the pooled sample only.

H The minimum-distance estimator

H.1 Moments, parameters, and the estimator

On the five waves of 2014 to 2022 the estimator uses the $K = 15$ pairwise autocovariances $\hat{\omega}_{ts}$, $t \leq s$, where t and s index waves. Each is the sample mean of $\tilde{y}_{it}\tilde{y}_{is}$ over the n_{ts} children observed in both waves, after demeaning within year (and within group for the fits by group). The model

(G.2) implies $\omega_{ts} = m_{s-t}(\theta)$ with $m_k(\theta) = \sigma_\eta^2 + \rho^k \sigma_\varepsilon^2 / (1 - \rho^2)$ and $\theta = (\sigma_\eta^2, \sigma_\varepsilon^2, \rho)$, so the model has $p = 3$ parameters and imposes $K - p = 12$ restrictions. The estimator weights each moment by its cell count n_{ts} and solves

$$\hat{\theta} = \arg \min_{\theta \in \Theta} Q_N(\theta), \quad Q_N(\theta) = \sum_{t \leq s} \frac{n_{ts}}{N} (\hat{\omega}_{ts} - m_{s-t}(\theta))^2, \quad J = N Q_N(\hat{\theta}), \quad (\text{H.1})$$

with $N = \sum_{t \leq s} n_{ts}$ and $\Theta = \{\sigma_\eta^2 \geq 0, \sigma_\varepsilon^2 > 0, |\rho| < 1\}$. This is minimum distance with the fixed diagonal weight matrix $W = \text{diag}(n_{ts}/N)$ (Chamberlain, 1984; Newey and McFadden, 1994).

Assumption H.1 (Sampling). Children are an i.i.d. draw; the pattern of observed waves is independent of $(\eta_i, \{\varepsilon_{it}\})$; and fourth moments of $\tilde{y}$ are finite.

Under Assumption H.1 and the model, $\sqrt{n}(\hat{\omega} - \omega(\theta_0)) \Rightarrow N(0, \Omega)$ for the vector of pairwise moments, where n is the number of children,

$$\Omega_{(ts), (t's')} = \frac{\pi_{ts, t's'}}{\pi_{ts} \pi_{t's'}} \left(E[\tilde{y}_t \tilde{y}_s \tilde{y}_{t'} \tilde{y}_{s'}] - \omega_{ts} \omega_{t's'} \right),$$

π_{ts} is the probability that a child is observed in both waves t and s , and $\pi_{ts, t's'}$ is the probability that a child is observed in all of the waves t, s, t' , and s' . The independence part of the assumption is what lets pairwise deletion estimate population autocovariances. It is a missing-at-random condition on wave participation. The screens that define the sample pass 45.5 percent of advantaged and 42.0 percent of less advantaged children (Table A3); the assumption concerns which waves a child in the sample is observed in, which the screens do not test.

H.2 Identification

Proposition H.1 (Identification from three lags). *Suppose the autocovariances at three consecutive lags $k_1, k_2 = k_1 + 1$, and $k_3 = k_1 + 2$ are known. On Θ with $\sigma_\varepsilon^2 > 0$, and with $\rho \neq 0$ if $k_1 \geq 1$, the parameter θ is identified:*

$$\begin{aligned} \rho &= \frac{\omega_{k_2} - \omega_{k_3}}{\omega_{k_1} - \omega_{k_2}}, \quad \sigma_v^2 = \frac{\omega_{k_1} - \omega_{k_2}}{\rho^{k_1}(1 - \rho)}, \\ \sigma_\eta^2 &= \omega_{k_1} - \rho^{k_1} \sigma_v^2, \quad \sigma_\varepsilon^2 = \sigma_v^2(1 - \rho^2). \end{aligned} \quad (\text{H.2})$$

With the variance among the three moments ($k_1 = 0$), identification fails only when $\sigma_\varepsilon^2 = 0$, in which case all autocovariances equal σ_η^2 and ρ is undetermined. It is lost in the limit $\rho \rightarrow 1$, where the autocovariance function becomes flat and the permanent variance cannot be separated from a transitory component that never decays.

Proof. From (G.2), $\omega_{k_1} - \omega_{k_2} = \rho^{k_1}(1 - \rho)\sigma_v^2$ and $\omega_{k_2} - \omega_{k_3} = \rho^{k_1+1}(1 - \rho)\sigma_v^2$. The denominator $\omega_{k_1} - \omega_{k_2}$ is nonzero when $\sigma_v^2 > 0$, $\rho \neq 1$, and either $k_1 = 0$ or $\rho \neq 0$, and the ratio is then ρ . The remaining expressions give σ_v^2 , σ_η^2 , and σ_ε^2 in turn. $\square$

The Jacobian $G = \partial\omega(\theta)/\partial\theta'$ has rows $(1, \rho^k/(1 - \rho^2), \sigma_\varepsilon^2 \partial_\rho[\rho^k/(1 - \rho^2)])$. The determinant of its three rows at lags 0, 1, and 2 is $\sigma_\varepsilon^2/(1 + \rho)^2$, so G has full column rank whenever $\sigma_\varepsilon^2 > 0$

and $|\rho| < 1$. The rank condition holds at every estimate in Table 3 of the paper, and the ridge of Section 3.3 of the paper is a matter of the conditioning of G and not of its rank. The next result quantifies the conditioning.

H.3 The identification ridge

Section 3.3 of the paper reports that σ_η^2 and ρ trade off along a ridge of the criterion, and that the bootstrap draws for advantaged children spread along it (Figure 2 of the paper). The following result describes that curve and shows why it flattens as persistence rises.

Proposition H.2 (Leverage of the higher-order autocorrelations along the level set of the first-order autocorrelation). *Write the model in autocorrelation form, $r_k = \omega_k/\omega_0 = s + (1-s)\rho^k$. Fix r_1 and move along the set of (s, ρ) pairs consistent with it, $\rho(s) = (r_1 - s)/(1 - s)$. Then the k -th autocorrelation changes along this set at the rate*

$$\left. \frac{dr_k}{ds} \right|_{r_1 \text{ fixed}} = h_k(\rho) \equiv 1 - \rho^k - k\rho^{k-1}(1 - \rho) = (1 - \rho)^2 \sum_{j=0}^{k-2} (j+1)\rho^j, \quad k \geq 2, \quad (\text{H.3})$$

so that $h_2 = (1 - \rho)^2$, $h_3 = (1 - \rho)^2(1 + 2\rho)$, and every h_k vanishes quadratically as $\rho \rightarrow 1$.

Proof. Differentiate $r_k(s) = s + (1-s)\rho(s)^k$ using $\rho'(s) = -(1 - r_1)/(1 - s)^2 = -(1 - \rho)/(1 - s)$: $dr_k/ds = 1 - \rho^k + (1-s)k\rho^{k-1}\rho'(s) = 1 - \rho^k - k\rho^{k-1}(1 - \rho)$. For the factorization, write $1 - \rho^k = (1 - \rho) \sum_{j=0}^{k-1} \rho^j$ and $k\rho^{k-1}(1 - \rho) = (1 - \rho) \sum_{j=0}^{k-1} \rho^{k-1-j}$, so $h_k = (1 - \rho) \sum_{j=0}^{k-1} (\rho^j - \rho^{k-1-j}) = (1 - \rho) \sum_{j=0}^{k-2} \rho^j (1 - \rho^{k-1-j})$, and expanding $1 - \rho^{k-1-j} = (1 - \rho) \sum_{i=0}^{k-2-j} \rho^i$ and collecting powers of ρ gives the stated sum. $\square$

The first-order autocorrelation fixes one combination of the permanent share and the persistence parameter. The autocorrelations of second and higher order separate the two, and they do so with leverage $h_k(\rho)$. At the estimate for less advantaged children, $\hat{\rho} = 0.12$, $h_2 = 0.77$: moving the permanent share by one unit along the level set of r_1 moves r_2 by 0.77, so the second-order moment determines the split tightly. At the estimate for advantaged children, $\hat{\rho} = 0.33$, $h_2 = 0.45$ and $h_3 = 0.75$. The leverage of the second-order moment is little more than half as large, and it is applied to moments from 420 children instead of 2,103, whose standard errors are larger by a factor of about 2.2 on sample size alone. Weaker leverage and noisier moments together produce the elongated cloud of bootstrap draws in Figure 2 of the paper. Proposition H.2 shows that the elongation is a property of the model at high ρ and not of the estimator: any method that identifies the permanent share from the decay of the autocovariance function faces the same $(1 - \rho)^2$ factor.

H.4 Asymptotic distribution and the choice of weights

Proposition H.3 (Asymptotics of the estimator with fixed weights). *Under Assumption H.1, if θ_0 is the unique minimizer of the population criterion on Θ (Proposition H.1), G has full column*

rank (shown above), θ_0 is interior to Θ , Ω is positive definite, and the weight matrix converges to a positive definite W ,

$$\sqrt{n}(\hat{\theta} - \theta_0) \Rightarrow N(0, (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}), \quad (\text{H.4})$$

and the criterion $J = NQ_N(\hat{\theta})$ converges in distribution to $\sum_{j=1}^{K-p} \nu_j \chi_{1,j}^2$, a weighted sum of independent χ_1^2 variables whose weights ν_j are the nonzero eigenvalues of $\Omega^{1/2}(W - WG(G'WG)^{-1}G'W)\Omega^{1/2}$ multiplied by the limit of N/n . For the optimally weighted criterion $n(\hat{\omega} - \omega(\theta))'\hat{\Omega}^{-1}(\hat{\omega} - \omega(\theta))$ the weights all equal one and the limit is χ_{K-p}^2 (Hansen, 1982). For the fixed weights of (H.1) the weights ν_j depend on Ω , and the χ_{K-p}^2 reference distribution does not apply.

Proof. The argument is the standard one for minimum distance (Newey and McFadden, 1994); consistency follows from their Theorem 2.1 on a compact subset of Θ that contains θ_0 in its interior, where the population criterion is continuous and has its unique minimum at θ_0 . The first-order condition $G'W(\hat{\omega} - \omega(\hat{\theta})) = 0$ and a mean-value expansion give (H.4). For the criterion, the minimized residual is $(I - G(G'WG)^{-1}G'W)(\hat{\omega} - \omega(\theta_0)) + o_p(n^{-1/2})$, a linear map of an asymptotically normal vector, so $J = (N/n) z' \Omega^{1/2}(W - WG(G'WG)^{-1}G'W)\Omega^{1/2}z + o_p(1)$ with z asymptotically standard normal. A quadratic form $z'Az$ in a standard normal vector with A symmetric is distributed as $\sum_j \nu_j \chi_{1,j}^2$ with ν_j the eigenvalues of A , and the matrix here has rank $K - p$. With $W = \Omega^{-1}$ it is a projection, whose nonzero eigenvalues equal one. $\square$

Section 3.3 of the paper reports J under fixed cell-count weights as a measure of fit and tests the model with a parametric bootstrap of the inverse-variance-weighted criterion. Proposition H.3 is the reason for both choices. The weights ν_j depend on Ω , and with fifteen pairwise moments from an unbalanced panel of 2,523 children the estimate $\hat{\Omega}$ is noisy. Altonji and Segal (1996) show that the optimally weighted estimator is biased in small samples because $\hat{\Omega}$ is correlated with the sample moments it weights. Fixed weights give up efficiency to avoid that bias. Inverse-variance reweighting of the moments moves the pooled permanent share from 0.271 to 0.264 (Section 3.3 of the paper), so the estimate is not sensitive to the choice between the two sets of weights.

H.5 Decomposing the criterion

The twelve overidentifying restrictions are of two kinds. Ten of them say that autocovariances at the same lag but different calendar dates are equal (stationarity). Two of them say that the five lag-specific autocovariances lie on the curve $\sigma_\eta^2 + \rho^k \sigma_v^2$ (shape). With a diagonal weight matrix the criterion splits exactly into the two.

Proposition H.4 (Stationarity and shape components of J). *Let $\mathcal{P}_k$ be the set of wave pairs at lag k , $W_k = \sum_{(t,s) \in \mathcal{P}_k} n_{ts}$, and $\bar{\omega}_k = W_k^{-1} \sum_{(t,s) \in \mathcal{P}_k} n_{ts} \hat{\omega}_{ts}$ the mean of the moments at lag k*

weighted by cell counts. Then for every θ ,

$$NQ_N(\theta) = \underbrace{\sum_k \sum_{(t,s) \in \mathcal{P}_k} n_{ts} (\hat{\omega}_{ts} - \bar{\omega}_k)^2}_{J_{\text{stat}}} + \underbrace{\sum_k W_k (\bar{\omega}_k - m_k(\theta))^2}_{J_{\text{shape}}(\theta)}, \quad (\text{H.5})$$

where J_{stat} does not depend on θ . Consequently the estimator on the fifteen pairwise moments coincides with the estimator that fits the five lag means with weights W_k , and $J = J_{\text{stat}} + J_{\text{shape}}(\hat{\theta})$ with $K - L = 10$ and $L - p = 2$ degrees of freedom attached to the two components, $L = 5$ being the number of distinct lags.

Proof. Within lag k , write $\hat{\omega}_{ts} - m_k = (\hat{\omega}_{ts} - \bar{\omega}_k) + (\bar{\omega}_k - m_k)$ and expand the weighted square; the cross term is $2(\bar{\omega}_k - m_k) \sum_{\mathcal{P}_k} n_{ts} (\hat{\omega}_{ts} - \bar{\omega}_k) = 0$ by the definition of the weighted mean. Summing over k gives (H.5). Since the first term is free of θ , minimizing Q_N is the same as minimizing J_{shape} . $\square$

Table A26: Decomposition of the minimum-distance criterion.

	J (12 df)	J_{stat} (10 df)	J_{shape} (2 df)	Stationarity share of J	Lag means $\bar{\omega}_0, \dots, \bar{\omega}_4$
Pooled	29.31	28.85	0.46	0.984	0.459, 0.177, 0.141, 0.123, 0.110
Advantaged children	2.66	2.51	0.14	0.946	0.438, 0.198, 0.128, 0.105, 0.066
Less advantaged children	29.76	29.51	0.25	0.992	0.460, 0.169, 0.141, 0.124, 0.120

Notes: Equation (H.5) at the minimum-distance estimates of Table 3 of the paper. J reproduces the paper's values (29.3, 2.7, 29.8). Degrees of freedom are nominal: by Proposition H.3 neither component is χ^2 under fixed weights, and the paper reports the total as a measure of fit only.

Table A26 reports the split. The stationarity component is 98.4 percent of the criterion in the pooled sample and 99.2 percent for less advantaged children. Autocovariances at the same lag differ across calendar dates, most visibly the variances, which are 0.58 in 2016 against 0.41 to 0.49 in the other waves (pooled row of Table A5). The shape component is 0.46 in the pooled sample: a permanent effect plus an AR(1) component reproduces the five lag means closely. The decomposition locates the misfit that the paper reports for the pooled sample and for less advantaged children. It lies in the departures of same-lag moments from equality across calendar dates, the sense in which the paper says that stationarity holds only approximately, and not in the shape of the curve. The estimates depend on the data only through the five lag means (Proposition H.4), so the variation across calendar dates does not enter them; when stationarity fails they describe the autocovariance function averaged over waves. For advantaged children the criterion is 2.66 and has the same composition, with 94.6 percent in the stationarity component. With 420 children, departures from stationarity of a given size add less to a criterion whose weights are cell counts.

H.6 Inference at the boundary

Remark H.5 (Bootstrap inconsistency at the boundary). Let $\hat{\theta}$ be the constrained estimator on Θ with $\sigma_\eta^2 \geq 0$. If $\sigma_{\eta,0}^2 = 0$, or if $\sqrt{n} \sigma_{\eta,0}^2$ is bounded, the limit distribution of $\sqrt{n}(\hat{\sigma}_\eta^2 - \sigma_{\eta,0}^2)$ is that of a normal variable censored from below, and the nonparametric bootstrap distribution of $\hat{\sigma}_\eta^2$ does not converge to it (Andrews, 2000). Percentile intervals for σ_η^2 , and for functions of it such as the permanent share and λ_T , are then not asymptotically valid. Andrews (2000) establishes the result for the mean of a normal sample restricted to be nonnegative, and the argument carries over: when the true parameter is on the boundary or within $O(n^{-1/2})$ of it, the bootstrap does not replicate the mass that the constrained estimator places at the boundary, because the bootstrap is centered at $\hat{\theta}$ and not at θ_0 .

The row of draws for advantaged children at a permanent share of zero in Figure 2 of the paper is this boundary. With $\hat{\sigma}_\eta^2 = 0.083$ and a bootstrap standard error of 0.028, the estimate for advantaged children is about three standard errors from the boundary, and the bootstrap draws reach it; this is the situation of Remark H.5. The bootstrap standard errors that Table 3 of the paper reports for that group carry this qualification, which is the reason Section 3.3 of the paper gives for not reading the ratio of a group gap to its standard error at face value. The paper treats the fitted split for advantaged children as suggestive and rests the comparison of groups on the fifteen raw moments (Figure 1 of the paper, Table A5, and the joint test of Section B.2). Their estimators are unconstrained sample means, to which the bootstrap applies without qualification.

H.7 Observed and transitory persistence

Under (G.1), the population first-order autocorrelation of $\tilde{y}$ is

$$\rho_{\text{obs}} = \text{Corr}(\tilde{y}_{it}, \tilde{y}_{i,t+1}) = s + (1-s)\rho, \quad (\text{H.6})$$

exactly: the ratio of $\text{Cov}(\tilde{y}_t, \tilde{y}_{t+1}) = \sigma_\eta^2 + \rho\sigma_v^2$ to $\text{Var}(\tilde{y}_t) = \sigma_\eta^2 + \sigma_v^2$, with s substituted, is (H.6). It is the probability limit of the pooled least squares coefficient of $\tilde{y}_{it}$ on $\tilde{y}_{i,t-1}$ without individual effects, which converges to the ratio of these two moments, and it exceeds ρ whenever $s > 0$.

Equation (3) of the paper states (H.6) under stationarity. In the population it is exact; applied to a pooled regression on an unbalanced panel whose waves have unequal variances (Table A5) it holds approximately. At the pooled estimates, $\rho_{\text{obs}} = 0.27 + 0.73 \times 0.16 = 0.39$, the value in Section 3.4 of the paper.

The Arellano–Bond estimator of Section 3.4 of the paper targets ρ and not ρ_{obs} . Rewriting (G.1) as $\tilde{y}_{it} = \rho\tilde{y}_{i,t-1} + (1-\rho)\eta_i + \varepsilon_{it}$ shows that the autoregression in equation (6) of the paper has individual effect $\alpha_i = (1-\rho)\eta_i$ and idiosyncratic error ε_{it} , which is serially uncorrelated, so lagged levels dated $t-2$ and earlier are valid instruments for the first-differenced equation (Arellano and Bond, 1991). The two estimators share a target, as the paper says, so the gap between the minimum-distance estimate and the Arellano–Bond estimate is not a difference of definitions; the paper notes that the minimum-distance estimate for less advantaged children, 0.12, lies inside the 95 percent interval of the Arellano–Bond estimate for that group, $[-0.01, 0.13]$.

I The accounting identity for the premium

Section 7 of the paper decomposes the premium through identity (1) of the paper, $\tau = \beta^H \Delta M + \text{Direct}$, where τ is the premium with parental income held fixed, ΔM is the gap in college completion between advantaged and less advantaged children, and β^H is the return to college for advantaged children. The paper evaluates the identity at the least squares returns and at the instrumented return to college. The following result states the identity in observed conditional means and shows what the direct term collects and what the remainder contains when the identity is evaluated at a return other than β^H . Expectations for less advantaged children are taken under the entropy-balancing weights of Section 5.3 of the paper, which match the first two moments of parental income, gender, age, urban status, five-year birth cohort, and province (the fifteen largest separately) to those of advantaged children.

Proposition I.1 (Content of the direct term). *Let $H \in \{0, 1\}$ indicate advantaged children and $M \in \{0, 1\}$ college completion. Define $p_h = \Pr(M = 1 \mid H = h)$, $a_h = E[Y \mid H = h, M = 0]$, the mean log earnings of the children of group h without a degree, and $r_h = E[Y \mid H = h, M = 1] - E[Y \mid H = h, M = 0]$, the earnings difference between the graduates of group h and its children without a degree. Then, with $\tau = E[Y \mid H = 1] - E[Y \mid H = 0]$, $\Delta M = p_1 - p_0$, and $\beta^H = r_1$ as in Section 7 of the paper:*

(i) *The identity holds with*

$$\tau = r_1 \Delta M + \text{Direct}, \quad \text{Direct} = (a_1 - a_0) + p_0(r_1 - r_0) = (a_1 + p_0 r_1) - (a_0 + p_0 r_0). \quad (\text{I.1})$$

The direct term is the sum of the gap between the groups in earnings without college and a term equal to the college completion rate of less advantaged children, p_0 , times the difference in returns between the two groups' graduates. It equals $a_1 - a_0$ if and only if $r_1 = r_0$ or $p_0 = 0$.

(ii) *For any number b ,*

$$\tau - b \Delta M = \text{Direct} + (r_1 - b) \Delta M. \quad (\text{I.2})$$

Proof. By the law of total expectation, $E[Y \mid H = h] = (1 - p_h) a_h + p_h (a_h + r_h) = a_h + p_h r_h$. Subtract the expression for $h = 0$ from the expression for $h = 1$ and add and subtract $p_0 r_1$: $\tau = (a_1 - a_0) + p_1 r_1 - p_0 r_0 = (a_1 - a_0) + (p_1 - p_0) r_1 + p_0(r_1 - r_0)$, which is (i). Part (ii) adds and subtracts $b \Delta M$. $\square$

The identity is an accounting identity in observed conditional means and requires no assumption. Written in potential outcomes, with $Y = Y(0) + M(Y(1) - Y(0))$, the same algebra holds with $a_h = E[Y(0) \mid H = h]$ and $r_h = E[Y(1) - Y(0) \mid H = h, M = 1]$; the observed r_h then equals that return plus the selection term $E[Y(0) \mid H = h, M = 1] - E[Y(0) \mid H = h, M = 0]$, which is why the paper reads the identity at a least squares return as accounting and not as a causal statement. The last expression in (I.1) gives the direct term the reading of Section 7 of the paper, the earnings advantage that would remain if the college gap were closed, in a specific

form: the completion rate of advantaged children is lowered from p_1 to p_0 , and the graduates removed have mean return r_1 . The components $\tau = 0.072$ and $\Delta M = 0.122$ are differences between the two groups after entropy balancing, and the paper reads the premium as an association. Inside the balanced comparison the identity is exact: graduates out-earn children without a degree by $r_1 = 0.541$ among advantaged children and by $r_0 = 0.517$ among reweighted less advantaged children, with $p_1 = 0.3355$ and $p_0 = 0.2130$, so $r_1 \Delta M = 0.066$, $p_0 (r_1 - r_0) = 0.005$, and $a_1 - a_0 = 0.001$ (Section 7 of the paper). A causal return takes the place of r_1 when the identity is evaluated at the instrument. The instrument gives an estimate of it for the advantaged children the expansion moved into college, 1.14, with a 90 percent Anderson–Rubin set that excludes every return between -3.20 and 0.60 (Table 11 of the paper), and an estimate for all children, 0.68, with a 95 percent Anderson–Rubin set that starts at 0.50 (Table 10 of the paper). The paper evaluates the identity over this range of returns.

At the least squares return, 0.449, the remainder is $0.072 - 0.449 \times 0.122 = 0.017$ (Table 12 of the paper); at the instrumented returns the college channel, 0.084 for all children and 0.140 for advantaged children, exceeds the premium. By (I.2) this remainder equals Direct if the least squares return equals r_1 . If the least squares return exceeds r_1 , the remainder understates Direct by 0.122 times the excess, and if it falls short of r_1 the remainder overstates Direct by 0.122 times the shortfall. Table 12 of the paper evaluates the left side of (I.2) at alternative values of b . The second term of Direct in (I.1) vanishes when returns among graduates are equal in the two groups. In the balanced comparison the two returns are 0.541 and 0.517, so the second term is 0.005 and the direct term exceeds the gap between the groups in earnings without a degree, $a_1 - a_0 = 0.001$, by that amount. When returns are higher for advantaged graduates ($r_1 > r_0$), Direct is an upper bound on $a_1 - a_0$.

Part III. Replication guide

J Replication files

J.1 Layout of the repository

The repository has five folders. Each pipeline folder holds its own code, its inputs, and the estimates it writes; the folder of the paper holds the scripts that combine them.

Table A27: Folders of the repository.

Folder	Content
unified	The paper (unified/paper); the scripts that build the sample of the paper and produce its descriptive facts, the premium, the expansion design, the decomposition, the tables, and every figure (unified/code); and their output (unified/output).
income_dynamics	Construction of the panel from the CFPS files, residualization, and the estimates of the earnings process and the attenuation schedule (Sections 3 and 4 of the paper).
college_expansion	Provincial intensity measures, the cohort exposure series, the placebo from examination registrations, the census check of trends before the reform, and the entropy-balancing routine that unified/code imports.
formal_results	Computations behind Part II of this appendix.
online_appendix	Source of this appendix, the script that writes its tables from the estimation output, and the checks of its numbers.

J.2 Data availability

Table A28 lists the data the paper uses and states for each whether the repository carries it and how to obtain it otherwise. The repository carries every aggregate series and every file of estimates. It does not carry person-level records from the China Family Panel Studies or the census samples, which their providers license to registered users and do not allow to be redistributed. A registered user who places the files in the folders named below can run the pipeline from the raw data.

Table A28: Data sources and access.

Data	In the repository	Access
China Family Panel Studies, waves 2010 to 2022 (adult, family roster, and family economy files)	No	Institute of Social Science Survey, Peking University. Registration and download at https://www.issp.pku.edu.cn/cfps/ . Place each wave in <code>income_dynamics/data/raw/CFPS2010, ... , CFPS2022</code> .
Census samples of 1990 and 2000 (one percent)	No	National Bureau of Statistics of China through IPUMS International (Ruggles et al., 2025), https://international.ipums.org . Variables: census year, age, province (<code>GE01_CN</code>), educational attainment, person weight. Save the extract in <code>college_expansion/data/external</code> as <code>ipumsi_00001.csv.gz</code> , the file that college_expansion/code/p16_census_pretrend.py reads.
Provincial intensity of the expansion	Yes	Constructed from the provincial series of Ma (2024): <code>college_expansion/data/external/expansion_intensity_real.csv</code> .
National admissions and the exposure ratio	Yes	<code>college_expansion/data/external/cohort_exposure.csv</code> .
Registrations for the college entrance examination by province, 1990s	Yes	<i>Chinese Educational Statistical Yearbook</i> : <code>college_expansion/data/external/cee_registrants_province.csv</code> .
Exports and GDP by province, 2000	Yes	<i>China Statistical Yearbook for Regional Economy</i> 2001: <code>college_expansion/data/external/expgdp2000.csv</code> .
Senior secondary completion of adults born 1956 to 1976, by province	Yes	Province-level shares computed from the census samples: <code>college_expansion/data/external/census_hs_stock.csv</code> .
Consumer prices by province and year	Yes	<code>college_expansion/data/external/cpi_province_year.csv</code> .

J.3 Software

The estimates are computed in Python 3.12 with `numpy`, `pandas`, `scipy`, `statsmodels`, `matplotlib`, and `pyreadstat`; the file [requirements.txt](#) lists the versions. The dynamic-panel estimates are also computed in Stata 18 with `xtabond2` ([income_dynamics/code/stata](#)). The paper and this appendix compile with pdfLaTeX and BibTeX.

J.4 Order of execution

1. *Panel and samples* (`./run_all.sh --from-raw`). [income_dynamics/code/01_build_panel.py](#) and [income_dynamics/code/02_sample_and_parents.py](#) read the CFPS files and link children to parents; [unified/code/d0_dynamics_panel.py](#) writes the child-wave panel of the children with at least three waves; [income_dynamics/code/03_step0_residualize.py](#) residualizes it, [income_dynamics/code/ws_panel_7wave.py](#) builds the

reconstructed seven-wave panel, and `income_dynamics/code/m14_parent_process.py` estimates the earnings process of the linked parents. These steps need the licensed files. The sample of the paper (children observed in at least two waves at ages 22 to 55), the level of earnings at ages 30 and over, and the two rows per child of the expansion design are built by `unified/code/sample_mature_age.py`, a module that the scripts of the later steps import.

2. *Earnings process and measurement.* The scripts listed in Table A29 for Sections 3 and 4 of the paper. At this step `run_all.sh` also runs seven scripts whose outputs the scripts of Table A29 and `unified/code/u7_error_structures.py` read: `income_dynamics/code/04_method1_md.py`, `income_dynamics/code/05_method2_abgmm.py`, `income_dynamics/code/06_mini_method3_randcoef.py`, `income_dynamics/code/ws_methods_7wave.py`, `income_dynamics/code/ws_method1_arma11.py`, `income_dynamics/code/ws_p1_6_ma1_rw.py`, and `income_dynamics/code/ws_m3_strict_exogeneity_test.py`.
3. *Premium.* `unified/code/u8_premium_specifications.py`, on the level of earnings at ages 30 and over; the balancing routine is that of `college_expansion/code/ws_07b_entropy_balance.py`.
4. *Expansion and decomposition.* `unified/code/u9_mature_age_earnings.py`, `unified/code/u12_returns_by_family_background.py`, and `unified/code/u13_decomposition_mature_age.py`, each run with unweighted rows and with rows weighted by the precision the earnings process implies. The first of these is also run for two blocks that are computed only on request: block G (the Anderson–Rubin p -value of the return for all children at every point of the grid, and the first stage of three-year completion) and block R (randomization inference). `unified/code/u14_sample_facts.py` describes the sample and supplies the first stages on all children; `unified/code/u16_identity_in_balanced_comparison.py` splits the premium inside the balanced comparison; `college_expansion/code/p16_census_pretrend.py` runs the census check; then the other scripts of `unified/code`, among them `unified/code/u17_income_variance_by_wave.py` and `unified/code/u15_nonlinear_persistence_runs.py`.
5. *Checks and documents.* `unified/code/check_paper_numbers.py` checks the numbers printed in the paper against the estimation output; `online_appendix/build/build_tables.py` writes the tables of Part I of this appendix; `online_appendix/build/check_part1_numbers.py` and `online_appendix/build/check_part2_numbers.py` check the numbers of Parts I and II.

The script `run_all.sh` executes steps 2 to 5 in this order, and step 1 as well when called with `--from-raw`. Most of the running time goes to the wild cluster bootstrap with 9,999 replications, the grid inversion behind the Anderson–Rubin sets, the randomization inference, and the ten runs of the stochastic estimator of nonlinear persistence.

J.5 Tables and figures of the paper

Table A29: The script and the output file behind each table and figure of the paper, and behind the numbers reported only in its text.

Exhibit of the paper	Script	Output
Table 2. The sample	<code>unified/code/u14_sample_facts.py</code>	<code>unified/output/u14_A_sample_means.csv</code>
Table 3. Permanent and transitory components	<code>income_dynamics/code/m7_md_cellweighted_boot.py</code>	<code>income_dynamics/output/tables/m7_md_cellw_boot.csv</code>
Figure 1. Variances and autocovariances by group	<code>unified/code/fig_autocov.py</code>	<code>unified/paper/figures/fig_autocov.png</code>
Figure 2. Identification ridge	<code>unified/code/fig_ridge.py</code>	<code>unified/output/u5_ridge_draws.csv</code>
Online Appendix Table A5. Autocovariances	<code>income_dynamics/code/m5_md_weighted_and_ridge.py</code>	<code>income_dynamics/output/tables/m5_moment_se.csv</code>
Table 4. Dynamic-panel estimates	<code>income_dynamics/code/m10_abgmm_table3.py</code>	<code>income_dynamics/output/tables/m10_abgmm_table3.csv</code>
Table 5. Comparison of groups	<code>income_dynamics/code/m13_mde_group_null.py</code> , <code>income_dynamics/code/m16_joint_moment_test.py</code> , <code>income_dynamics/code/m10_abgmm_table3.py</code> , <code>income_dynamics/code/ws_05b_abb_stochastic_em.py</code> , <code>unified/code/u10_growth_gap.py</code>	<code>income_dynamics/output/tables/m13_mde.csv</code> , <code>unified/output/u10_growth_gap.csv</code>
Table 6. Attenuation schedule	<code>formal_results/code/appendix_theory_computations.py</code>	<code>formal_results/output/A_eta_bootstrap.csv</code>
Table 6, panel B. Corrected published estimates	<code>unified/code/u2_recovery_by_process.py</code>	<code>unified/output/u2_recovery_by_process.csv</code>
Table 7. The premium	<code>unified/code/u8_premium_specifications.py</code> , <code>college_expansion/code/ws_07b_entropy_balance.py</code>	<code>unified/output/u8_premium_specifications.csv</code> , <code>unified/output/u8_entropy_balancing.csv</code>
Table 8. Competing markers	<code>unified/code/u8_premium_specifications.py</code>	<code>unified/output/u8_premium_specifications.csv</code>
Figure 3. Fifteen specifications	<code>unified/code/u8_premium_specifications.py</code> , <code>unified/code/fig_spec_curve.py</code>	<code>unified/output/u8_premium_specifications.csv</code>
Table 9. Assumptions, threats, and diagnostics	<code>unified/code/u9_mature_age_earnings.py</code> , <code>unified/code/u12_returns_by_family_background.py</code> , <code>unified/code/u13_decomposition_mature_age.py</code> , <code>college_expansion/code/p16_census_pretrend.py</code>	<code>unified/output/u9_B_main.csv</code> , <code>unified/output/u9_C_checks.csv</code> , <code>unified/output/u12_returns_by_family_background.csv</code> , <code>unified/output/u13_C_design_facts.csv</code>
Figure 4. College completion and first stage	<code>unified/code/fig_first_stage.py</code>	<code>unified/paper/figures/fig_first_stage.png</code> , <code>unified/output/fig_first_stage_left.csv</code>

Exhibit of the paper	Script	Output
Table 10. Earnings and occupation at ages 30 and over; first stages with one observation per child	<code>unified/code/u9_mature_age_earnings.py</code> , <code>unified/code/u14_sample_facts.py</code>	<code>unified/output/u9_B_main.csv</code> , <code>unified/output/u14_C_first_stages_all_children.csv</code>
Figure 5. Event study	<code>unified/code/u9_mature_age_earnings.py</code>	<code>unified/output/u9_D_cohort_bins.csv</code> , <code>unified/paper/figures/fig_mature_age.png</code>
Table 11. Returns for all children and by family background	<code>unified/code/u12_returns_by_family_background.py</code> , <code>unified/code/u9_mature_age_earnings.py</code>	<code>unified/output/u12_returns_by_family_background.csv</code> , <code>unified/output/u12_anderson_rubin_grid.csv</code> , <code>unified/output/u9_B_main.csv</code> , <code>unified/output/u9_anderson_rubin_grid.csv</code>
Table 12. The direct effect; the three parts of the premium inside the balanced comparison	<code>unified/code/u13_decomposition_mature_age.py</code> , <code>unified/code/u16_identity_in_balanced_comparison.py</code>	<code>unified/output/u13_B_identity.csv</code> , <code>unified/output/u16_identity_in_balanced_comparison.csv</code>
<i>Numbers reported in the text of the paper and not in a table or figure</i>		
Section 2.1. Variance of log income by wave	<code>unified/code/u17_income_variance_by_wave.py</code>	<code>unified/output/u17_income_variance_by_wave.csv</code>
Section 2.2. Screens, employment, least squares return by age	<code>unified/code/u14_sample_facts.py</code>	<code>unified/output/u14_B_sample_facts.csv</code> , <code>unified/output/u14_D_return_by_age.csv</code>
Section 3.3. Criterion values under alternative error structures; inverse-variance reweighting and its parametric bootstrap	<code>unified/code/u7_error_structures.py</code> , <code>income_dynamics/code/m5_md_weighted_and_ridge.py</code>	<code>unified/output/u7_error_structures.csv</code> , <code>income_dynamics/output/tables/m5_weighted_md.csv</code>
Section 3.5. Test of strict exogeneity	<code>income_dynamics/code/ws_m3_strict_exogeneity_test.py</code>	<code>income_dynamics/data/intermediate/ws/ws_m3_strict_exogeneity.csv</code>
Section 3.5. Nonlinear persistence, ten runs	<code>income_dynamics/code/ws_05b_abb_stochastic_em.py</code> , <code>unified/code/u15_nonlinear_persistence_runs.py</code>	<code>unified/output/u15_nonlinear_persistence_summary.csv</code>
Section 6.2. Randomization inference	<code>unified/code/u9_mature_age_earnings.py</code> (block R)	<code>unified/output/u9_R_randomization_inference.csv</code> , <code>unified/output/u9_R_randomization_within_region.csv</code>
Section 6.4. Three-year completion; set from randomization inference	<code>unified/code/u9_mature_age_earnings.py</code> (blocks G and R)	<code>unified/output/u9_G_three_year_completion.csv</code> , <code>unified/output/u9_R_randomization_within_region.csv</code>

Table 1 of the paper contains no estimates beyond those of the exhibits listed here. The tables of

this appendix name their scripts in their notes.

J.6 How the tables of estimates are produced

The eight tables of the paper that report means or estimates with standard errors (Tables 2, 3, 4, 5, 7, 8, 10 and 11) are generated rather than typed. The script `unified/code/build_paper_tables.py` reads the files of estimates named in its header, formats each coefficient with its significance stars and standard error, and writes the body of each table to `unified/paper/tables/en/`; the source of the paper reads these fragments. The differential-growth estimate of Table 5 comes from `unified/code/u10_growth_gap.py`. Running the script with the option `--check` compares the fragments on disk with what the files of estimates imply, and the check described below runs that comparison.

J.7 How the printed numbers are checked

The estimates printed in the paper are checked against the files of estimates. For each of them, `unified/code/check_paper_numbers.py` reads the file of estimates, formats the value as it is printed, and searches for that string in the source of the paper; it fails if the string is absent. The script also checks the arithmetic the text performs on estimates, such as the share of the gap that a correction closes or the magnitude implied by a difference in the instrument. The two checks of this appendix work the same way.

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